CHAPTER 43 SUMMARY

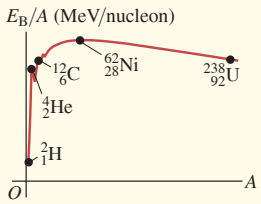
Nuclear properties: A nucleus is composed of A nucleons (Z protons and N neutrons). All nuclei have about the same density. The radius of a nucleus with mass number A is given approximately by Eq. (43.1). A single nuclear species of a given Z and N is called a nuclide. Isotopes are nuclides of the same element (same Z) that have different numbers of neutrons. Nuclear masses are measured in atomic mass units. Nucleons have angular momentum and a magnetic moment. (See Examples 43.1 and 43.2.)

$$R = R_0 A^{1/3} \qquad (43.1)$$

$$(R_0 = 1.2 \times 10^{-15}\ \text{m})$$

Nuclear binding and structure: The mass of a nucleus is always less than the mass of the protons and neutrons within it. The mass difference multiplied by c^2 gives the binding energy E_B. The binding energy for a given nuclide is determined by the nuclear force, which is short range and favors pairs of particles, and by the electrical repulsion between protons. A nucleus is unstable if A or Z is too large or if the ratio N/Z is wrong. Two widely used models of the nucleus are the liquid-drop model and the shell model; the latter is analogous to the central-field approximation for atomic structure. (See Examples 43.3 and 43.4.)

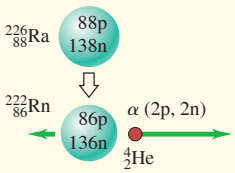
$$E_B = (ZM_H + Nm_n - {}^A_Z M)c^2 \qquad (43.10)$$



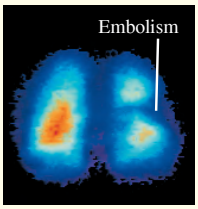
Radioactive decay: Unstable nuclides usually emit an alpha particle (a $^4_2$He nucleus) or a beta particle (an electron) in the process of changing to another nuclide, sometimes followed by a gamma-ray photon. The rate of decay of an unstable nucleus is described by the decay constant λ, the half-life $T_{1/2}$, or the lifetime T_{mean}. If the number of nuclei at time $t = 0$ is N_0 and no more are produced, the number at time t is given by Eq. (43.17). (See Examples 43.5–43.9.)

$$N(t) = N_0 e^{-\lambda t} \qquad (43.17)$$

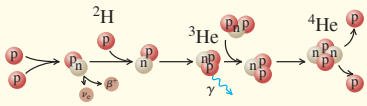
$$T_{mean} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693} \qquad (43.19)$$



Biological effects of radiation: The biological effect of any radiation depends on the product of the energy absorbed per unit mass and the relative biological effectiveness (RBE), which is different for different radiations. (See Example 43.10.)



Nuclear reactions: In a nuclear reaction, two nuclei or particles collide to produce two new nuclei or particles. Reactions can be exoergic or endoergic. Several conservation laws, including charge, energy, momentum, angular momentum, and nucleon number, are obeyed. Energy is released by the fission of a heavy nucleus into two lighter, always unstable, nuclei. Energy is also released by the fusion of two light nuclei into a heavier nucleus. (See Examples 43.11–43.13.)



Chapter 43 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 43.3 and 43.4 (Section 43.2) before attempting these problems.

VP43.4.1 Find (a) the mass defect, (b) the total binding energy, and (c) the binding energy per nucleon of $^{16}_{8}O$, which has a neutral atomic mass of 15.994915 u.

VP43.4.2 The mass defect of $^{63}_{29}Cu$ is 0.5919378 u. Find (a) its binding energy, (b) its binding energy per nucleon, and (c) its neutral atomic mass.

VP43.4.3 Calculate the binding energy E_B/A for the nuclides (a) $^{75}_{33}As$ (mass defect 0.7005604 u), (b) $^{150}_{62}Sm$ (mass defect 1.330388 u), and (c) $^{225}_{88}Ra$ (mass defect 1.852094 u). (d) Figure 43.2 shows that for nuclei with nucleon numbers $A > 62$, the binding energy per nucleon E_B/A decreases as A increases. Do your results agree with this?

VP43.4.4 Use the semiempirical mass formula to calculate the total estimated binding energy E_B and the binding energy per nucleon E_B/A for the nuclides (a) ruthenium-100 ($^{100}_{44}Ru$) and (b) mercury-200 ($^{200}_{80}Hg$). (c) Are your results consistent with Fig. 43.2?

Be sure to review EXAMPLES 43.5, 43.6, and 43.7 (Section 43.3) before attempting these problems.

VP43.7.1 Household smoke detectors contain a tiny amount of radioactive americium that decays by α emission to neptunium: $^{241}_{95}Am \rightarrow {}^{237}_{93}Np + {}^{4}_{2}He$. The atomic masses are 241.056827 u for $^{241}_{95}Am$, 237.048172 u for $^{237}_{93}Np$, and 4.002603 u for $^{4}_{2}He$. If the $^{241}_{95}Am$ nucleus is originally at rest, find (a) the total energy released in the decay, (b) the kinetic energy of the emitted α particle, and (c) the speed of the emitted α particle.

VP43.7.2 The beryllium isotope $^{8}_{4}Be$ is very unstable, and rapidly decays into two α particles: $^{8}_{4}Be \rightarrow {}^{4}_{2}He + {}^{4}_{2}He$. The atomic masses are 8.0053051 u for $^{8}_{4}Be$ and 4.0026033 u for $^{4}_{2}He$. If the $^{8}_{4}Be$ nucleus is originally at rest, find (a) the kinetic energy and (b) the speed of each emitted α particle.

VP43.7.3 State whether each of the following proposed β^- decays is possible or impossible. If it is possible, find the total energy released in the decay. The atomic mass is given for each nuclide. (a) $^{12}_{4}Be$ (12.026922 u) to $^{12}_{5}B$ (12.014353 u); (b) $^{33}_{17}Cl$ (32.977452 u) to $^{33}_{18}Ar$ (32.989926 u); (c) $^{82}_{35}Br$ (81.9168018 u) to $^{82}_{36}Kr$ (81.9134812 u).

VP43.7.4 State whether each of the following proposed β^+ decays is possible or impossible. If it is possible, find the total energy released in the decay; if it is impossible, state whether the decay can occur by electron capture. The atomic mass is given for each nuclide. The combined mass of two electrons is 0.001097 u. (a) $^{46}_{23}V$ (45.960198 u) to $^{46}_{22}Ti$ (45.952627 u); (b) $^{134}_{57}La$ (133.908514 u) to $^{134}_{56}Ba$ (133.904508 u); (c) $^{67}_{31}Ga$ (66.9282024u) to $^{67}_{30}Zn$ (66.9271275 u).

Be sure to review EXAMPLES 43.8 and 43.9 (Section 43.4) before attempting these problems.

VP43.9.1 The sodium isotope $^{24}_{11}Na$ undergoes β^- decay to $^{24}_{12}Mg$ (a stable isotope) with a half-life of 15.0 h. Initially a sample of $^{24}_{11}Na$ has β^- activity of 1.50 μCi. Find (a) the mean lifetime, (b) the decay constant, (c) the initial number of $^{24}_{11}Na$ nuclei in the sample, and (d) the activity of the sample after 24.0 h.

VP43.9.2 You are given a sample that was initially 100% $^{95}_{41}Nb$. This isotope undergoes β^- decay to $^{95}_{42}Mo$ (a stable isotope) with a half-life of 35.0 h. You find that the sample now contains 23.8% $^{95}_{41}Nb$ and 76.2% $^{95}_{42}Mo$ and has a β^- activity of 0.514 μCi. Find (a) the decay constant, (b) the present number of $^{95}_{41}Nb$ nuclei in the sample, (c) the initial number of $^{95}_{41}Nb$ nuclei in the sample, and (d) how much time has elapsed since the sample was 100% $^{95}_{41}Nb$.

VP43.9.3 A certain isotope undergoes γ decay from an excited state to its ground state. Initially a sample of this isotope has an activity of 0.425 μCi; 45.0 s later, the activity has decreased to 0.121 μCi. Find (a) the decay constant and half-life of the excited state and (b) the number of nuclei in the excited state initially and 45.0 s later.

VP43.9.4 Before 1900 the activity per unit mass of atmospheric carbon due to the presence of ^{14}C (half-life 5730 y) averaged about 0.255 Bq per gram of carbon. You detect 113 decays of ^{14}C in 20.0 min from a sample of carbon of mass 0.510 g. Assuming that its activity per unit mass of carbon when it formed was the pre-1900 average value for the air, find (a) the age of the sample and (b) the current number of ^{14}C nuclei in the sample.

BRIDGING PROBLEM Saturation of ^{128}I Production

In an experiment, the iodine isotope ^{128}I is created by irradiating a sample of ^{127}I with a beam of neutrons, yielding 1.50×10^6 ^{128}I nuclei per second. Initially no ^{128}I nuclei are present. A ^{128}I nucleus decays by β^- emission with a half-life of 25.0 min. (a) To what nuclide does ^{128}I decay? (b) Could that nuclide decay back to ^{128}I by β^+ emission? Why or why not? (c) After the sample has been irradiated for a long time, what is the maximum number of ^{128}I atoms that can be present in the sample? What is the maximum activity that can be produced? (This steady-state situation is called *saturation*.) (d) Find an expression for the number of ^{128}I atoms present in the sample as a function of time.

SOLUTION GUIDE

IDENTIFY and SET UP

1. What happens to the values of Z, N, and A in β^- decay? What must be true for β^- decay to be possible? For β^+ decay to be possible?
2. You'll need to write an equation for the rate of change dN/dt of the number N of ^{128}I atoms in the sample, taking account of both the creation of ^{128}I by the neutron irradiation and the decay of any ^{128}I present. In the steady state, how do the rates of these two processes compare?
3. List the unknown quantities for each part of the problem and identify your target variables.

EXECUTE

4. Find the values of Z and N of the nuclide produced by the decay of ^{128}I. What element is this?
5. Decide whether this nuclide can decay back to ^{128}I.
6. Inspect your equation for dN/dt. What is the value of dN/dt in the steady state? Use this to solve for the steady-state values of N and the activity.
7. Solve your dN/dt equation for the function $N(t)$. (*Hint:* See Section 26.4.)

EVALUATE

8. Your result from step 6 tells you the value of N after a long time (that is, for large values of t). Is this consistent with your result from step 7? What would constitute a "long time" under these conditions?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q43.1 **BIO** Neutrons have a magnetic dipole moment and can undergo spin flips by absorbing electromagnetic radiation. Why, then, are protons rather than neutrons used in MRI of body tissues? (See Fig. 43.1.)

Q43.2 In Eq. (43.11), as the total number of nucleons becomes larger, the importance of the second term in the equation decreases relative to that of the first term. Does this make physical sense? Explain.

Q43.3 Why aren't the masses of all nuclei integer multiples of the mass of a single nucleon?

Q43.4 The only two stable nuclides with more protons than neutrons are ${}^{1}_{1}\text{H}$ and ${}^{3}_{2}\text{He}$. Why is $Z > N$ so uncommon?

Q43.5 What are the six known elements for which Z is a magic number? Discuss what properties these elements have as a consequence of their special values of Z.

Q43.6 The binding energy per nucleon for most nuclides doesn't vary much (see Fig. 43.2). Is there similar consistency in the *atomic* energy of atoms, on an "energy per electron" basis? If so, why? If not, why not?

Q43.7 Heavy, unstable nuclei usually decay by emitting an α or a β particle. Why don't they usually emit a single proton or neutron?

Q43.8 As stars age, they use up their supply of hydrogen and eventually begin producing energy by a reaction that involves the fusion of three helium nuclei to form a carbon nucleus. Would you expect the interiors of these old stars to be hotter or cooler than the interiors of younger stars? Explain.

Q43.9 Since lead is a stable element, why doesn't the ${}^{238}\text{U}$ decay series shown in Fig. 43.7 stop at lead, ${}^{214}\text{Pb}$?

Q43.10 In the ${}^{238}\text{U}$ decay series shown in Fig. 43.7, some nuclides in the series are found much more abundantly in nature than others, even though every ${}^{238}\text{U}$ nucleus goes through every step in the series before finally becoming ${}^{206}\text{Pb}$. Why don't the intermediate nuclides all have the same abundance?

Q43.11 Compared to α particles with the same energy, β particles can much more easily penetrate through matter. Why is this?

Q43.12 If ${}^{A}_{Z}\text{El}_\text{i}$ represents the initial nuclide, what is the decay process or processes if the final nuclide is (a) ${}_{Z+1}^{\ \ A}\text{El}_\text{f}$; (b) ${}^{A-4}_{Z-2}\text{El}_\text{f}$; (c) ${}_{Z-1}^{\ \ A}\text{El}_\text{f}$?

Q43.13 In a nuclear decay equation, why can we represent an electron as ${}^{0}_{-1}\beta^-$? What are the equivalent representations for a positron, a neutrino, and an antineutrino?

Q43.14 Why is the alpha, beta, or gamma decay of an unstable nucleus unaffected by the *chemical* situation of the atom, such as the nature of the molecule or solid in which it is bound? The chemical situation of the atom can, however, have an effect on the half-life in electron capture. Why is this?

Q43.15 In the process of *internal conversion,* a nucleus decays from an excited state to a ground state by giving the excitation energy directly to an atomic electron rather than emitting a gamma-ray photon. Why can this process also produce x-ray photons?

Q43.16 In Example 43.9 (Section 43.4), the activity of atmospheric carbon *before* 1900 was given. Discuss why this activity may have changed since 1900.

Q43.17 **BIO** One problem in radiocarbon dating of biological samples, especially very old ones, is that they can easily be contaminated with modern biological material during the measurement process. What effect would such contamination have on the estimated age? Why is such contamination a more serious problem for samples of older material than for samples of younger material?

Q43.18 The most common radium isotope found on earth, ${}^{226}\text{Ra}$, has a half-life of about 1600 years. If the earth was formed well over 10^9 years ago, why is there any radium left now?

Q43.19 Fission reactions occur only for nuclei with large nucleon numbers, while exoergic fusion reactions occur only for nuclei with small nucleon numbers. Why is this?

Q43.20 When a large nucleus splits during nuclear fission, the daughter nuclei of the fission fly apart with enormous kinetic energy. Why does this happen?

EXERCISES

Section 43.1 Properties of Nuclei

43.1 • How many protons and how many neutrons are there in a nucleus of the most common isotope of (a) silicon, ${}^{28}_{14}\text{Si}$; (b) rubidium, ${}^{85}_{37}\text{Rb}$; (c) thallium, ${}^{205}_{81}\text{Tl}$?

43.2 • Hydrogen atoms are placed in an external magnetic field. The protons can make transitions between states in which the nuclear spin component is parallel and antiparallel to the field by absorbing or emitting a photon. What magnetic-field magnitude is required for this transition to be induced by photons with frequency 23.5 MHz?

Section 43.2 Nuclear Binding and Nuclear Structure

43.3 • The most common isotope of boron is ${}^{11}_{5}\text{B}$. (a) Determine the total binding energy of ${}^{11}_{5}\text{B}$ from Table 43.2 in Section 43.1. (b) Calculate this binding energy from Eq. (43.11). (Why is the fifth term zero?) Compare to the result you obtained in part (a). What is the percent difference? Compare the accuracy of Eq. (43.11) for ${}^{11}_{5}\text{B}$ to its accuracy for ${}^{62}_{28}\text{Ni}$ (see Example 43.4).

43.4 •• The nuclei ${}^{11}_{5}\text{B}$ and ${}^{11}_{6}\text{C}$ are called *mirror nuclei,* because the number of protons in ${}^{11}_{5}\text{B}$ equals the number of neutrons in ${}^{11}_{6}\text{C}$ and the number of neutrons in ${}^{11}_{5}\text{B}$ equals the number of protons in ${}^{11}_{6}\text{C}$. The atomic mass of ${}^{11}_{6}\text{C}$ is 11.011434 u, and the atomic mass of ${}^{11}_{5}\text{B}$ is given in Table 43.2. (a) Calculate the binding energies of ${}^{11}_{5}\text{B}$ and ${}^{11}_{6}\text{C}$. (b) Which of the two nuclei has the larger binding energy? Why is this so?

43.5 • Calculate (a) the total binding energy and (b) the binding energy per nucleon of ${}^{12}\text{C}$. (c) What percent of the rest mass of this nucleus is its total binding energy?

43.6 • The most common isotope of uranium, ${}^{238}_{92}\text{U}$, has atomic mass 238.050788 u. Calculate (a) the mass defect; (b) the binding energy (in MeV); (c) the binding energy per nucleon.

43.7 • The binding energy per nucleon for ${}^{56}_{26}\text{Fe}$ is 8.79 MeV/nucleon. What is the mass, in amu, of a neutral ${}^{56}_{26}\text{Fe}$ atom? Give your answer to five significant figures.

43.8 •• An alpha particle is strongly bound. The ${}^{12}_{6}\text{C}$ nucleus might be modeled as a composite of three alpha particles. Compare the binding energy of ${}^{12}_{6}\text{C}$ with three times the binding energy of an alpha particle. Which of these quantities is larger, and why might this be so?

43.9 • **CP** A photon with a wavelength of 3.00×10^{-13} m strikes a deuteron, splitting it into a proton and a neutron. (a) Calculate the kinetic energy released in this interaction. (b) Assuming the two particles share the energy equally, and taking their masses to be 1.00 u, calculate their speeds after the photodisintegration.

43.10 • Calculate the mass defect, the binding energy (in MeV), and the binding energy per nucleon of (a) the nitrogen nucleus, ${}^{14}_{7}\text{N}$, and (b) the helium nucleus, ${}^{4}_{2}\text{He}$. (c) How does the binding energy per nucleon compare for these two nuclei?

Section 43.3 Nuclear Stability and Radioactivity

43.11 • The carbon isotope $^{11}_{6}C$ undergoes β^+ (positron) decay. The atomic mass of $^{11}_{6}C$ is 11.011433 u. (a) How many protons and how many neutrons are in the daughter nucleus produced by this decay? (b) How much energy, in MeV, is released in the decay of one $^{11}_{6}C$ nucleus?

43.12 • (a) Is the decay $n \rightarrow p + \beta^- + \bar{\nu}_e$ energetically possible? If not, explain why not. If so, calculate the total energy released. (b) Is the decay $p \rightarrow n + \beta^+ + \nu_e$ energetically possible? If not, explain why not. If so, calculate the total energy released.

43.13 • What nuclide is produced in the following radioactive decays? (a) α decay of $^{239}_{94}Pu$; (b) β^- decay of $^{24}_{11}Na$; (c) β^+ decay of $^{15}_{8}O$.

43.14 •• CP ^{238}U decays spontaneously by α emission to ^{234}Th. Calculate (a) the total energy released by this process and (b) the recoil velocity of the ^{234}Th nucleus. The atomic masses are 238.050788 u for ^{238}U and 234.043601 u for ^{234}Th.

43.15 • The radioactive isotope $^{12}_{7}N$ undergoes β^+ decay. The energy released in the decay of one nucleus is 16.316 MeV. What is the mass, in amu, of a neutral $^{12}_{7}N$ atom? Give your answer to eight significant figures.

43.16 • What particle (α particle, electron, or positron) is emitted in the following radioactive decays? (a) $^{27}_{14}Si \rightarrow ^{27}_{13}Al$; (b) $^{238}_{92}U \rightarrow ^{234}_{90}Th$; (c) $^{74}_{33}As \rightarrow ^{74}_{34}Se$.

43.17 • Tritium ($^{3}_{1}H$) is an unstable isotope of hydrogen; its mass, including one electron, is 3.016049 u. (a) Show that tritium must be unstable with respect to beta decay because the decay products ($^{3}_{2}He$ plus an emitted electron) have less total mass than the tritium. (b) Determine the total kinetic energy (in MeV) of the decay products, taking care to account for the electron masses correctly.

Section 43.4 Activities and Half-Lives

43.18 • A sample of radioactive nuclei has N_0 nuclei at time $t = 0$. The half-life of the decay is $T_{1/2}$. In terms of N_0, how many decays occur in the time period between $t = 0$ and $t = 0.500T_{1/2}$?

43.19 • If a 5.63 g sample of an isotope having a mass number of 114 decays at a rate of 0.400 Ci, what is its half-life?

43.20 • BIO Radioactive isotopes used in cancer therapy have a "shelf-life," like pharmaceuticals used in chemotherapy. Just after it has been manufactured in a nuclear reactor, the activity of a sample of ^{60}Co is 5000 Ci. When its activity falls below 3500 Ci, it is considered too weak a source to use in treatment. You work in the radiology department of a large hospital. One of these ^{60}Co sources in your inventory was manufactured on October 6, 2011. It is now April 6, 2014. Is the source still usable? The half-life of ^{60}Co is 5.271 years.

43.21 •• The common isotope of uranium, ^{238}U, has a half-life of 4.47×10^9 years, decaying to ^{234}Th by alpha emission. (a) What is the decay constant? (b) What mass of uranium is required for an activity of 1.00 curie? (c) How many alpha particles are emitted per second by 10.0 g of uranium?

43.22 •• BIO **Radiation Treatment of Prostate Cancer.** In many cases, prostate cancer is treated by implanting 60 to 100 small seeds of radioactive material into the tumor. The energy released from the decays kills the tumor. One isotope that is used (there are others) is palladium (^{103}Pd) with a half-life of 17 days. If a typical grain contains 0.250 g of ^{103}Pd, (a) what is its initial activity rate in Bq, and (b) what is the rate 68 days later?

43.23 •• A 12.0 g sample of carbon from living matter decays at the rate of 172.5 decays/minute due to the radioactive ^{14}C in it. What will be the decay rate of this sample in (a) 1000 years and (b) 50,000 years?

43.24 •• BIO **Radioactive Tracers.** Radioactive isotopes are often introduced into the body through the bloodstream. Their spread through the body can then be monitored by detecting the appearance of radiation in different organs. One such tracer is ^{131}I, a β^- emitter with a half-life of 8.0 d. Suppose a scientist introduces a sample with an activity of 325 Bq and watches it spread to the organs. (a) Assuming that all of the sample went to the thyroid gland, what will be the decay rate in that gland 24 d (about $3\frac{1}{2}$ weeks) later? (b) If the decay rate in the thyroid 24 d later is measured to be 17.0 Bq, what percentage of the tracer went to that gland? (c) What isotope remains after the I-131 decays?

43.25 • The isotope $^{90}_{39}Y$ undergoes β^- decay with a half-life of 64.0 h. You measure an activity of 8.0×10^{16} Bq. (a) How many $^{90}_{39}Y$ nuclei are present in the sample at the time you make this measurement? (b) How many will be present after 12.0 days?

43.26 • As a health physicist, you are being consulted about a spill in a radiochemistry lab. The isotope spilled was 400 μCi of ^{131}Ba, which has a half-life of 12 days. (a) What mass of ^{131}Ba was spilled? (b) Your recommendation is to clear the lab until the radiation level has fallen 1.00 μCi. How long will the lab have to be closed?

43.27 • Measurements on a certain isotope tell you that the decay rate decreases from 8265 decays/min to 3088 decays/min in 4.00 days. What is the half-life of this isotope?

43.28 •• **Radiocarbon Dating.** At an archeological site, a sample from timbers containing 500 g of carbon provides 2690 decays/min. What is the age of the sample?

43.29 • The radioactive nuclide ^{199}Pt has a half-life of 30.8 minutes. A sample is prepared that has an initial activity of 7.22×10^{11} Bq. (a) How many ^{199}Pt nuclei are initially present in the sample? (b) How many are present after 30.8 minutes? What is the activity at this time? (c) Repeat part (b) for a time 92.4 minutes after the sample is first prepared.

Section 43.5 Biological Effects of Radiation

43.30 • BIO **Radiation Overdose.** If a person's entire body is exposed to 5.0 J/kg of x rays, death usually follows within a few days. (a) Express this lethal radiation dose in Gy, rad, Sv, and rem. (b) How much total energy does a 70.0 kg person absorb from such a dose? (c) If the 5.0 J/kg came from a beam of protons instead of x rays, what would be the answers to parts (a) and (b)?

43.31 •• BIO A nuclear chemist receives an accidental radiation dose of 4.0 Gy from slow neutrons (RBE = 4.0). What does she receive in rad, rem, and J/kg?

43.32 •• BIO A person exposed to fast neutrons receives a radiation dose of 300 rem on part of his hand, affecting 25 g of tissue. The RBE of these neutrons is 10. (a) How many rad did he receive? (b) How many joules of energy did he receive? (c) Suppose the person received the same rad dosage, but from beta rays with an RBE of 1.0 instead of neutrons. How many rem would he have received?

43.33 • BIO **Food Irradiation.** Food is often irradiated with either x rays or electron beams to help prevent spoilage. A low dose of 5–75 kilorads (krad) helps to reduce and kill inactive parasites, a medium dose of 100–400 krad kills microorganisms and pathogens such as salmonella, and a high dose of 2300–5700 krad sterilizes food so that it can be stored without refrigeration. (a) A dose of 175 krad kills spoilage microorganisms in fish. If x rays are used, what would be the dose in Gy, Sv, and rem, and how much energy would a 250 g portion of fish absorb? (See Table 43.3.) (b) Repeat part (a) if electrons of RBE 1.50 are used instead of x rays.

43.34 • BIO **To Scan or Not to Scan?** It has become popular for some people to have yearly whole-body scans (CT scans, formerly called CAT scans) using x rays, just to see if they detect anything suspicious. A number of medical people have recently questioned the advisability of such scans, due in part to the radiation they impart. Typically, one such scan gives a dose of 12 mSv, applied to the *whole body*. By contrast, a chest x ray typically administers 0.20 mSv to only 5.0 kg of tissue. How many chest x rays would deliver the same *total* amount of energy to the body of a 75 kg person as one whole-body scan?

43.35 •• The radioactive isotope $^{90}_{38}Sr$ is found in the fallout from atmospheric tests of nuclear weapons (see Problem 43.49). It replaces calcium in items such as bone and milk. It has a half-life of 29 years, and the net energy absorbed from each decay averages about 1.1 MeV. (a) Calculate the absorbed dose from 1.0 microgram of $^{90}_{38}Sr$ in one year if the energy is absorbed in 50 kg of body tissue. (b) The radiation from $^{90}_{38}Sr$ involves beta-minus particles and gamma rays. Calculate the equivalent dose in part (a) if you assume the RBE for the electrons is 1.0.

43.36 • **BIO** In an industrial accident a 61 kg person receives a lethal whole-body equivalent dose of 6.0 Sv from x rays. (a) What is the equivalent dose in rem? (b) What is the absorbed dose in rad? (c) What is the total energy absorbed by the person's body? How does this amount of energy compare to the amount of energy required to raise the temperature of 61 kg of water 0.010° C?

43.37 •• **CP** **BIO** In a diagnostic x-ray procedure, 5.00×10^{10} photons are absorbed by tissue with a mass of 0.600 kg. The x-ray wavelength is 0.0200 nm. (a) What is the total energy absorbed by the tissue? (b) What is the equivalent dose in rem?

Section 43.6 Nuclear Reactions

Section 43.7 Nuclear Fission

Section 43.8 Nuclear Fusion

43.38 •• Calculate the reaction energy Q for the reaction $p + {}^{3}_{1}H \rightarrow {}^{2}_{1}H + {}^{2}_{1}H$. Is this reaction exoergic or endoergic?

43.39 • Consider the nuclear reaction

$$^{2}_{1}H + {}^{9}_{4}Be \rightarrow X + {}^{4}_{2}He$$

where X is a nuclide. (a) What are the values of Z and A for the nuclide X? (b) How much energy is liberated? (c) Estimate the threshold energy for this reaction.

43.40 • **Energy from Nuclear Fusion.** Calculate the energy released in the fusion reaction

$$^{3}_{2}He + {}^{2}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{1}H$$

43.41 •• At the beginning of Section 43.7 the equation of a fission process is given in which ^{235}U is struck by a neutron and undergoes fission to produce ^{144}Ba, ^{89}Kr, and three neutrons. The measured masses of these isotopes are 235.043930 u (^{235}U), 143.922953 u (^{144}Ba), 88.917631 u (^{89}Kr), and 1.0086649 u (neutron). (a) Calculate the energy (in MeV) released by each fission reaction. (b) Calculate the energy released per gram of ^{235}U, in MeV/g.

43.42 •• Consider the nuclear reaction

$$^{28}_{14}Si + \gamma \rightarrow {}^{24}_{12}Mg + X$$

where X is a nuclide. (a) What are Z and A for the nuclide X? (b) Ignoring the effects of recoil, what minimum energy must the photon have for this reaction to occur? The mass of a $^{28}_{14}Si$ atom is 27.976927 u, and the mass of a $^{24}_{12}Mg$ atom is 23.985042 u.

43.43 • Consider the nuclear reaction

$$^{4}_{2}He + {}^{7}_{3}Li \rightarrow X + {}^{1}_{0}n$$

where X is a nuclide. (a) What are Z and A for the nuclide X? (b) Is energy absorbed or liberated? How much?

PROBLEMS

43.44 •• **CP** The nucleus $^{10}_{4}Be$ undergoes β^- decay. The atomic mass of $^{10}_{4}Be$ is 10.013535 u. (a) How much energy, in MeV, is released in this decay? (b) If you assume that all the energy released in the decay appears as kinetic energy of the β^- particle (a very accurate assumption), what is the speed of the emitted β^-? Note that the energy released is nearly the same as the rest energy of the β^-.

43.45 • **Comparison of Energy Released per Gram of Fuel.** (a) When petrol is burned, it releases 3.45×10^7 J of energy per liter. Given that the density of petrol is 737 kg/m^3, express the quantity of energy released in J/g of fuel. (b) During fission, when a neutron is absorbed by a ^{235}U nucleus, about 200 MeV of energy is released for each nucleus that undergoes fission. Express this quantity in J/g of fuel. (c) In the proton-proton chain that takes place in stars like our sun, the overall fusion reaction can be summarized as six protons fusing to form one ^{4}He nucleus with two leftover protons and the liberation of 26.7 MeV of energy. The fuel is the six protons. Express the energy produced here in units of J/g of fuel. Notice the huge difference between the two forms of nuclear energy, on the one hand, and the chemical energy from petrol, on the other. (d) Our sun produces energy at a measured rate of 3.86×10^{26} W. If its mass of 1.99×10^{30} kg were all petrol, how long could it last before consuming all its fuel? (*Historical note:* Before the discovery of nuclear fusion and the vast amounts of energy it releases, scientists were confused. They knew that the earth was at least many millions of years old, but could not explain how the sun could survive that long if its energy came from chemical burning.)

43.46 •• (a) Calculate the minimum energy required to remove one proton from the nucleus $^{12}_{6}C$. This is called the proton-removal energy. (*Hint:* Find the difference between the mass of a $^{12}_{6}C$ nucleus and the mass of a proton plus the mass of the nucleus formed when a proton is removed from $^{12}_{6}C$.) (b) How does the proton-removal energy for $^{12}_{6}C$ compare to the binding energy per nucleon for $^{12}_{6}C$, calculated using Eq. (43.10)?

43.47 •• (a) Calculate the minimum energy required to remove one neutron from the nucleus $^{17}_{8}O$. This is called the neutron-removal energy. (See Problem 43.46.) (b) How does the neutron-removal energy for $^{17}_{8}O$ compare to the binding energy per nucleon for $^{17}_{8}O$, calculated using Eq. (43.10)?

43.48 •• Heat flows to the surface of the earth from its interior. Two-thirds of this heat is radiogenic in origin, which means it comes from the decay of radioactive elements in the earth's mantle, primarily from the isotopes ^{238}U and ^{232}Th. We can estimate the earth's internal energy supply due to the decay of uranium as follows: (a) There are 31 μg of ^{238}U in each kg of mantle. Estimate how many ^{238}U isotopes there are per kg of mantle. (b) The half-life of ^{238}U is 4.47 billion years. Calculate the decay constant in units of s^{-1}. (c) Multiply the results in parts (a) and (b) to obtain the activity in each kg of mantle. (d) Each decay chain $^{238}U \rightarrow {}^{206}Pb$ releases 52 MeV of energy. Use this value to convert the activity into watts per kg of mantle. (e) The mass of the earth is 6×10^{24} kg, and two-thirds of this is mantle mass. Use these values to estimate the power earth receives from uranium decays. (f) Uranium decays account for 39% of the radiogenic supply of heat. Estimate earth's total heat power related to radioactive decays. (g) Earth also releases heat that remains from its formation, with a power comparable with that due to radioactivity. Estimate the total heat power earth receives from its interior.

43.49 • **BIO** **Radioactive Fallout.** One of the problems of in-air testing of nuclear weapons (or, even worse, the *use* of such weapons!) is the danger of radioactive fallout. One of the most problematic nuclides in such fallout is strontium-90 (^{90}Sr), which breaks down by β^- decay with a half-life of 28 years. It is chemically similar to calcium and therefore can be incorporated into bones and teeth, where, due to its rather long half-life, it remains for years as an internal source of radiation. (a) What is the daughter nucleus of the ^{90}Sr decay? (b) What percentage of the original level of ^{90}Sr is left after 56 years? (c) How long would you have to wait for the original level to be reduced to 0.024% of its original value?

43.50 •• **CP** Thorium $^{230}_{90}Th$ decays to radium $^{226}_{88}Ra$ by α emission. The masses of the neutral atoms are 230.033134 u for $^{230}_{90}Th$ and 226.025410 u for $^{226}_{88}Ra$. If the parent thorium nucleus is at rest, what is the kinetic energy of the emitted α particle? (Be sure to account for the recoil of the daughter nucleus.)

43.51 •• The atomic mass of $^{25}_{12}$Mg is 24.985837 u, and the atomic mass of $^{25}_{13}$Al is 24.990428 u. (a) Which of these nuclei will decay into the other? (b) What type of decay will occur? Explain how you determined this. (c) How much energy (in MeV) is released in the decay?

43.52 •• The fusion reactions in the sun produce 26.73 MeV per alpha particle created. Each of these reactions produces two neutrinos. In total, the sun's power is about 3.8×10^{26} W. (a) Estimate how many solar fusion reactions take place every second. (b) Estimate the number of neutrinos that leave the sun each second. (c) The distance between the earth and the sun is 150 million km, and the radius of the earth is 6370 km. Using these figures, estimate the number of solar neutrinos that encounter the earth each second. (d) How many square centimeters does the earth present to the sun? (e) Divide your answer in part (c) by your answer in part (d) to obtain an estimate of the number of solar neutrinos that pass through every square centimeter on earth every second.

43.53 •• **BIO** **Irradiating Ourselves!** The radiocarbon in our bodies is one of the naturally occurring sources of radiation. Let's see how large a dose we receive. ^{14}C decays via β^- emission, and 18% of our body's mass is carbon. (a) Write out the decay scheme of carbon-14 and show the end product. (A neutrino is also produced.) (b) Neglecting the effects of the neutrino, how much kinetic energy (in MeV) is released per decay? The atomic mass of ^{14}C is 14.003242 u. (c) How many grams of carbon are there in a 75 kg person? How many decays per second does this carbon produce? (*Hint:* Use data from Example 43.9.) (d) Assuming that all the energy released in these decays is absorbed by the body, how many MeV/s and J/s does the ^{14}C release in this person's body? (e) Consult Table 43.3 and use the largest appropriate RBE for the particles involved. What radiation dose does the person give himself in a year, in Gy, rad, Sv, and rem?

43.54 •• **BIO** **Pion Radiation Therapy.** A neutral pion (π^0) has a mass of 264 times the electron mass and decays with a lifetime of 8.4×10^{-17} s to two photons. Such pions are used in the radiation treatment of some cancers. (a) Find the energy and wavelength of these photons. In which part of the electromagnetic spectrum do they lie? What is the RBE for these photons? (b) If you want to deliver a dose of 200 rem (which is typical) in a single treatment to 25 g of tumor tissue, how many π^0 mesons are needed?

43.55 •• Calculate the mass defect for the β^+ decay of $^{11}_{6}$C. Is this decay energetically possible? Why or why not? The atomic mass of $^{11}_{6}$C is 11.011434 u.

43.56 •• **BIO** A person ingests an amount of a radioactive source that has a very long lifetime and activity 0.72 μCi. The radioactive material lodges in her lungs, where all of the emitted 4.0 MeV α particles are absorbed within a 0.51 kg mass of tissue. Calculate the absorbed dose and the equivalent dose for one year.

43.57 • **We Are Stardust.** In 1952 spectral lines of the element technetium-99 (^{99}Tc) were discovered in a red giant star. Red giants are very old stars, often around 10 billion years old, and near the end of their lives. Technetium has *no* stable isotopes, and the half-life of ^{99}Tc is 200,000 years. (a) For how many half-lives has the ^{99}Tc been in the red giant star if its age is 10 billion years? (b) What fraction of the original ^{99}Tc would be left at the end of that time? This discovery was extremely important because it provided convincing evidence for the theory (now essentially known to be true) that most of the atoms heavier than hydrogen and helium were made inside of stars by thermonuclear fusion and other nuclear processes. If the ^{99}Tc had been part of the star since it was born, the amount remaining after 10 billion years would have been so minute that it would not have been detectable. This knowledge is what led the late astronomer Carl Sagan to proclaim that "we are stardust."

43.58 • **BIO** A 66.0 kg person experiences a whole-body exposure to α radiation with energy 4.72 MeV. A total of 6.30×10^{12} α particles are absorbed. (a) What is the absorbed dose in rad? (b) What is the equivalent dose in rem? (c) If the source is 0.0324 g of ^{226}Ra (half-life 1600 y) somewhere in the body, what is the activity of this source? (d) If all of the alpha particles produced are absorbed, what time is required for this dose to be delivered?

43.59 •• Measurements indicate that 27.83% of all rubidium atoms currently on the earth are the radioactive ^{87}Rb isotope. The rest are the stable ^{85}Rb isotope. The half-life of ^{87}Rb is 4.75×10^{10} y. Assuming that no rubidium atoms have been formed since, what percentage of rubidium atoms were ^{87}Rb when our solar system was formed 4.6×10^9 y ago?

43.60 • The nucleus $^{15}_{8}$O has a half-life of 122.2 s; $^{19}_{8}$O has a half-life of 26.9 s. If at some time a sample contains equal amounts of $^{15}_{8}$O and $^{19}_{8}$O, what is the ratio of $^{15}_{8}$O to $^{19}_{8}$O (a) after 3.0 min and (b) after 12.0 min?

43.61 •• **BIO** A ^{60}Co source with activity 2.2×10^{-4} Ci is embedded in a tumor that has mass 0.570 kg. The source emits γ photons with average energy 1.25 MeV. Half the photons are absorbed in the tumor, and half escape. (a) What energy is delivered to the tumor per second? (b) What absorbed dose (in rad) is delivered per second? (c) What equivalent dose (in rem) is delivered per second if the RBE for these γ rays is 0.70? (d) What exposure time is required for an equivalent dose of 195 rem?

43.62 ••• The proton-proton chain described in Section 43.8, known as the p-p I chain, accounts for 83.3% of the helium synthesis in the sun. This process uses six protons to produce one alpha particle, two protons, two neutrinos, and a least six gamma-ray photons. We can determine the energies of each of the products as follows: (a) The first reaction, $2\,^1_1\text{H} \rightarrow\, ^2_1\text{H} + \beta^+ + \upsilon_e$, is followed immediately by the annihilation of the positron by an electron into two photons, each with energy equal to the rest energy of an electron. What is the remaining energy E_1 carried by the deuteron and the neutrino? (b) Work in the frame where the net momentum of the deuteron and neutrino is zero. Denote the (relativistic) energy of the neutrino by E_υ and the (nonrelativistic) deuteron speed by υ_d. Denote the mass of the deuteron by m_d. What are the conditions for energy and momentum conservation for these two particles, in terms of E_1? Treat the neutrino as massless. (c) Solve the equations from part (b) and use the deuteron mass in Table 43.2 to determine the percentage of the energy E_1 carried by the neutrino. (d) What is the total energy carried away by neutrinos in the p-p I chain? (*Note:* This chain creates two neutrinos by the above mechanism.) (e) What percentage of the energy generated by the p-p I chain is carried away by neutrinos? (*Note:* The total energy of the p-p I chain is computed in Section 43.8.)

43.63 ••• The sun produces a minority 16.68% of its helium-4 using the p-p II chain, in which four protons and an ambient alpha particle produce two alpha particles, two neutrinos, and at least four gamma-ray photons. The first two steps are the same as the first two steps of the p-p I chain (see Problem 43.62). However, the third step is $^3_2\text{He} + \,^4_2\text{He} \rightarrow\, ^7_4\text{Be} + \gamma$, where $^7_4$Be is an unstable beryllium nuclide with mass 7.016930 u. (a) What is the energy of the photon? (b) The beryllium nuclide then captures an electron, which changes one of its protons into a neutron, resulting in $^7_4\text{Be} + e^- \rightarrow\, ^7_3\text{Li} + \upsilon_e$, where $^7_3$Li is the common (and stable) nuclide of lithium, with mass 7.016003 u. The excess energy is carried away by the neutrino. What is the energy of that neutrino? (c) A proton then collides with the lithium nucleus, resulting in two alpha particles via $^1_1\text{H} + \,^7_3\text{Li} \rightarrow 2\,^4_2\text{He}$. How much energy is released in this step? (d) If each alpha particle has the same speed, then in the rest frame of the lithium disintegration, what is the speed of each alpha particle? (e) How much energy is released in total by the processes in this p-p II chain? (f) How does this compare with the energy released in the p-p I chain? (g) What percentage of the energy in the p-p II chain is carried away by neutrinos?

43.64 ••• In addition to the p-p I and p-p II chains (see Problems 43.62 and 43.63), the sun produces helium-4 in a relatively rare reaction called the p-p III chain. This produces only 0.1% of the sun's alpha particles but results in neutrinos with substantially higher energy, which have proved crucial to the study of neutrino physics. In the p-p III chain, an ambient ${}^{3}_{2}\text{He}$ fuses with ambient ${}^{4}_{2}\text{He}$ (alpha particles) to produce ${}^{7}_{4}\text{Be}$, precisely as in the p-p II chain. But in the p-p III chain, the ${}^{7}_{4}\text{Be}$ then absorbs a proton to produce ${}^{8}_{5}\text{B}$. The ${}^{8}_{5}\text{B}$ then decays to ${}^{8}_{4}\text{Be}$ along with a positron and a neutrino. (a) The total energy released in this process can be ascertained from its net input and output products $4\,{}^{1}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He}$. How much energy is released? (b) Given the isotope masses 8.024607 u for ${}^{8}_{5}\text{B}$ and 8.0053051 u for ${}^{8}_{4}\text{Be}$, determine the energy of the neutrino emitted in the ${}^{8}_{5}\text{B}$ to ${}^{8}_{4}\text{Be}$ decay. (c) What percentage of the energy in the p-p III chain is carried away by neutrinos? (*Note:* This chain also produces a low-energy neutrino.)

43.65 • A bone fragment found in a cave believed to have been inhabited by early humans contains 0.21 times as much ${}^{14}\text{C}$ as an equal amount of carbon in the atmosphere when the organism containing the bone died. (See Example 43.9 in Section 43.4.) Find the approximate age of the fragment.

43.66 •• **BIO** In the 1986 disaster at the Chernobyl reactor in Ukraine, about $\frac{1}{8}$ of the ${}^{137}\text{Cs}$ present in the reactor was released. The isotope ${}^{137}\text{Cs}$ has a half-life of 30.07 y for β decay, with the emission of a total of 1.17 MeV of energy per decay. Of this, 0.51 MeV goes to the emitted electron; the remaining 0.66 MeV goes to a γ ray. The radioactive ${}^{137}\text{Cs}$ is absorbed by plants, which are eaten by livestock and humans. How many ${}^{137}\text{Cs}$ atoms would need to be present in each kilogram of body tissue if an equivalent dose for one week is 3.5 Sv? Assume that all of the energy from the decay is deposited in 1.0 kg of tissue and that the RBE of the electrons is 1.5.

43.67 •• Consider the fusion reaction ${}^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{2}\text{He} + {}^{1}_{0}\text{n}$. (a) Estimate the barrier energy by calculating the repulsive electrostatic potential energy of the two ${}^{2}_{1}\text{H}$ nuclei when they touch. (b) Compute the energy liberated in this reaction in MeV and in joules. (c) Compute the energy liberated *per mole* of deuterium, remembering that the gas is diatomic, and compare with the heat of combustion of hydrogen, about 2.9×10^5 J/mol.

43.68 •• **DATA** As a scientist in a nuclear physics research lab, you are conducting a photodisintegration experiment to verify the binding energy of a deuteron. A photon with wavelength λ in air is absorbed by a deuteron, which breaks apart into a neutron and a proton. The two fragments share the released kinetic energy equally, and the deuteron can be assumed to be initially at rest. You measure the speed of the proton after the disintegration as a function of the wavelength λ of the photon. Your experimental results are given in the table.

λ (10^{-13} m)	3.50	3.75	4.00	4.25	4.50	4.75	5.00
v (10^6 m/s)	11.3	10.2	9.1	8.1	7.2	6.1	4.9

(a) Graph the data as v^2 versus $1/\lambda$. Explain why the data points, when graphed this way, should follow close to a straight line. Find the slope and y-intercept of the straight line that gives the best fit to the data. (b) Assume that h and c have their accepted values. Use your results from part (a) to calculate the mass of the proton and the binding energy (in MeV) of the deuteron.

43.69 •• **DATA** Your company develops radioactive isotopes for medical applications. In your work there, you measure the activity of a radioactive sample. Your results are given in the table.

Time (h)	Decays/s
0	20,000
0.5	14,800
1.0	11,000
1.5	8130
2.0	6020
2.5	4460
3.0	3300
4.0	1810
5.0	1000
6.0	550
7.0	300

(a) Find the half-life of the sample. (b) How many radioactive nuclei were present in the sample at $t = 0$? (c) How many were present after 7.0 h?

43.70 ••• **DATA** In your job as a health physicist, you measure the activity of a mixed sample of radioactive elements. Your results are given in the table.

Time (h)	Decays/s
0	7500
0.5	4120
1.0	2570
1.5	1790
2.0	1350
2.5	1070
3.0	872
4.0	596
5.0	404
6.0	288
7.0	201
8.0	140
9.0	98
10.0	68
12.0	33

(a) What minimum number of different nuclides are present in the mixture? (b) What are their half-lives? (c) How many nuclei of each type are initially present in the sample? (d) How many of each type are present at $t = 5.0$ h?

CHALLENGE PROBLEMS

43.71 ••• **Industrial Radioactivity.** Radioisotopes are used in a variety of manufacturing and testing techniques. Wear measurements can be made using the following method. An automobile engine is produced using piston rings with a total mass of 100 g, which includes 9.4 μCi of ${}^{59}\text{Fe}$ whose half-life is 45 days. The engine is test-run for 1000 hours, after which the oil is drained and its activity is measured. If the activity of the engine oil is 84 decays/s, how much mass was worn from the piston rings per hour of operation?

43.72 ••• Many radioactive decays occur within a sequence of decays—for example, $^{234}_{92}U \rightarrow ^{230}_{88}Th \rightarrow ^{226}_{84}Ra$. The half-life for the $^{234}_{92}U \rightarrow ^{230}_{88}Th$ decay is 2.46×10^5 y, and the half-life for the $^{230}_{88}Th \rightarrow ^{226}_{84}Ra$ decay is 7.54×10^4 y. Let 1 refer to $^{234}_{92}U$, 2 to $^{230}_{88}Th$, and 3 to $^{226}_{84}Ra$; let λ_1 be the decay constant for the $^{234}_{92}U \rightarrow ^{230}_{88}Th$ decay and λ_2 be the decay constant for the $^{230}_{88}Th \rightarrow ^{226}_{84}Ra$ decay. The amount of $^{230}_{88}Th$ present at any time depends on the rate at which it is produced by the decay of $^{234}_{92}U$ and the rate by which it is depleted by its decay to $^{226}_{84}Ra$. Therefore, $dN_2(t)/dt = \lambda_1 N_1(t) - \lambda_2 N_2(t)$. If we start with a sample that contains N_{10} nuclei of $^{234}_{92}U$ and nothing else, then $N(t) = N_{01}e^{-\lambda_1 t}$. Thus $dN_2(t)/dt = \lambda_1 N_{10}e^{-\lambda_1 t} - \lambda_2 N_2(t)$. This differential equation for $N_2(t)$ can be solved as follows. Assume a trial solution of the form $N_2(t) = N_{10}[h_1 e^{-\lambda_1 t} + h_2 e^{-\lambda_2 t}]$, where h_1 and h_2 are constants. (a) Since $N_2(0) = 0$, what must be the relationship between h_1 and h_2? (b) Use the trial solution to calculate $dN_2(t)/dt$, and substitute that into the differential equation for $N_2(t)$. Collect the coefficients of $e^{-\lambda_1 t}$ and $e^{-\lambda_2 t}$. Since the equation must hold at all t, each of these coefficients must be zero. Use this requirement to solve for h_1 and thereby complete the determination of $N_2(t)$. (c) At time $t = 0$, you have a pure sample containing 30.0 g of $^{234}_{92}U$ and nothing else. What mass of $^{230}_{88}Th$ is present at time $t = 2.46 \times 10^5$ y, the half-life for the $^{234}_{92}U$ decay?

MCAT-STYLE PASSAGE PROBLEMS

BIO Radioactive Iodine in Medicine. Iodine in the body is preferentially taken up by the thyroid gland. Therefore, radioactive iodine in small doses is used to image the thyroid and in large doses is used to kill thyroid cells to treat some types of cancer or thyroid disease. The iodine isotopes used have relatively short half-lives, so they must be produced in a nuclear reactor or accelerator. One isotope frequently used for imaging is ^{123}I; it has a half-life of 13.2 h and emits a 0.16 MeV gamma-ray photon. One method of producing ^{123}I is in the nuclear reaction $^{123}Te + p \rightarrow ^{123}I + n$. The atomic masses relevant to this reaction are ^{123}Te, 122.904270 u; ^{123}I, 122.905589 u; n, 1.008665 u; and ^{1}H, 1.007825 u.

The iodine isotope commonly used for treatment of disease is ^{131}I, which is produced by irradiating ^{130}Te in a nuclear reactor to form ^{131}Te. The ^{131}Te then decays to ^{131}I. ^{131}I undergoes β^- decay with a half-life of 8.04 d, emitting electrons with energies up to 0.61 MeV and gamma-ray photons of energy 0.36 MeV. A typical thyroid cancer treatment might involve administration of 3.7 GBq of ^{131}I.

43.73 Which reaction produces ^{131}Te in the nuclear reactor? (a) $^{130}Te + n \rightarrow ^{131}Te$; (b) $^{130}I + n \rightarrow ^{131}Te$; (c) $^{132}Te + n \rightarrow ^{131}Te$; (d) $^{132}I + n \rightarrow ^{131}Te$.

43.74 Which type of radioactive decay produces ^{131}I from ^{131}Te? (a) Alpha decay; (b) β^- decay; (c) β^+ decay; (d) gamma decay.

43.75 How many ^{131}I atoms are administered in a typical thyroid cancer treatment? (a) 4.2×10^{10}; (b) 1.0×10^{12}; (c) 2.5×10^{14}; (d) 3.7×10^{15}.

43.76 In the reaction that produces ^{123}I, is there a minimum kinetic energy the protons need to make the reaction go? (a) No, because the proton has a smaller mass than the neutron. (b) No, because the total initial mass is smaller than the total final mass. (c) Yes, because the proton has a smaller mass than the neutron. (d) Yes, because the total initial mass is smaller than the total final mass.

43.77 Why might ^{123}I be preferred for imaging over ^{131}I? (a) The atomic mass of ^{123}I is smaller, so the ^{123}I particles travel farther through tissue. (b) Because ^{123}I emits only gamma-ray photons, the radiation dose to the body is lower with that isotope. (c) The beta particles emitted by ^{131}I can leave the body, whereas the gamma-ray photons emitted by ^{123}I cannot. (d) ^{123}I is radioactive, whereas ^{131}I is not.

ANSWERS

Chapter Opening Question ?

(iv) When an organism dies, it stops taking in carbon from atmospheric CO_2. Some of this carbon is radioactive ^{14}C, which decays with a half-life of 5730 years. By measuring the proportion of ^{14}C that remains in the specimen, scientists can determine how long ago the organism died. (See Section 43.4.)

Key Example VARIATION Problems

VP43.4.1 (a) 0.137005 u (b) 127.6 MeV (c) 7.976 MeV
VP43.4.2 (a) 551.4 MeV (b) 8.752 MeV (c) 62.929597 u
VP43.4.3 (a) 8.701 MeV (b) 8.262 MeV (c) 7.668 MeV (d) yes
VP43.4.4 (a) $E_B = 868.1$ MeV, $E_B/A = 8.681$ MeV (b) $E_B = 1584$ MeV, $E_B/A = 7.922$ MeV (c) yes
VP43.7.1 (a) 5.637 MeV (b) 5.543 MeV (c) 1.63×10^7 m/s
VP43.7.2 (a) 0.0459 MeV = 45.9 keV (b) 1.49×10^6 m/s
VP43.7.3 (a) possible, 11.71 MeV (b) impossible (c) possible, 3.093 MeV
VP43.7.4 (a) possible, 6.030 MeV (b) possible, 2.710 MeV (c) impossible, yes
VP43.9.1 (a) 7.79×10^4 s (b) $1.28 \times 10^{-5}\,s^{-1}$ (c) 4.32×10^9 (d) 0.495 μCi
VP43.9.2 (a) $5.50 \times 10^{-6}\,s^{-1}$ (b) 3.46×10^9 (c) 1.45×10^{10} (d) 72.5 h
VP43.9.3 (a) $\lambda = 0.0279\,s^{-1}$, $T_{1/2} = 24.8$ s (b) 5.63×10^5 initially, 1.60×10^5 after 45.0 s
VP43.9.4 (a) 2.67×10^3 y (b) 2.46×10^{10}

Bridging Problem

(a) ^{128}Xe
(b) no; β^+ emission would be endoergic
(c) 3.25×10^9 atoms, 1.50×10^6 Bq
(d) $N(t) = (3.25 \times 10^9 \text{ atoms})(1 - e^{-(4.62\times10^{-4}\,s^{-1})t})$